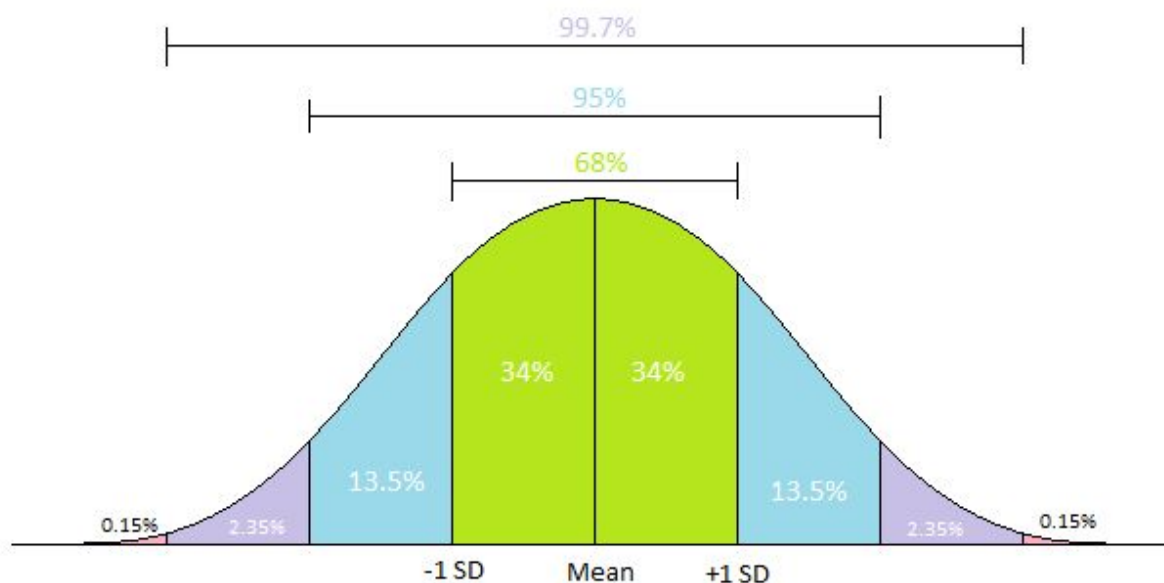




Probability



Module Leader Enikő Dinnyés

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Structure 8 credits, one 50-minute long class per week for 8 weeks

Module Summary

The module covers major topics of probability theory and statistics (up to first year university level). We will introduce theoretical concepts through simple experiments and games. Then we discuss some important probability distributions: what kind of distribution shall we use for the lifetime of light bulbs, or for the number of mistypings in pages of a newspaper, etc. How can we describe a probability distribution: distribution functions and density functions (and also characteristic functions). Law of large numbers in a mathematical formulation. Central limit theorem - or why the Gaussian (Normal) distribution is special. These concepts are used in financial mathematics. The above concepts form the basis of statistics: giving tools for appraising the world around us. Estimates, hypothesis testing. These are used in social sciences as well.

The module will be useful especially for those preparing to write STEP papers (applying to Cambridge or Imperial College London), because choosing questions from the Probability and Statistics part is an option there. We are going to solve some STEP questions in the module, too.

Module Pre-requirements

For completing the module, students need a firm mathematical background (module Calculus 1 or equivalent knowledge is preferable). Not so much lexical knowledge, but strong mathematical thinking is required.

Schedule

There will be 8 classes in total. The following topics will be covered during the course:

1. Event sets, combinatorial examples. Expectation of a game.
2. Independence of events, conditional probability, Bayes' theorem.
3. Distributions. Distribution function, probability function (discrete random variables), density function (continuous random variables).
4. Important discrete distributions (Geometric, Binomial, Poisson). Expectation, variance.
5. Important continuous distributions (Uniform, Normal, Exponential).
6. Transformation of a Normal distribution to Standard Normal. Generating function, characteristic function; sum of independent random variables.
7. Central Limit Theorem, Law of Large Numbers.
8. Statistical hypothesis tests.

Learning Outcomes

Many of your day-to-day assumptions already depend on statistics. You can improve this ability into more rigorous thinking: getting to more accurate conclusions, based on observations about our world.

Problem solving ability will also be improved, especially useful for those preparing to write STEP papers where Probability and Statistics is a separate topic.

Assessment

There are 4 problem sheets that must be solved and submitted during the course, after every second class. Each problem sheet contains 10 exercises, and each of these exercises are worth 10 points.

Grading

Each assignment (problem sheet) has a 25% weight in grading the course. At the end of the course, the average of the achievements in the four problem sheets is calculated as a percentage. The grading scheme works on a 1-10 scale with 1, 2 or 3 being a fail and 10 being the best grade awarded. The grade boundaries are 95%, 85%, 75%, 65%, 55%, 45%, 35%, 25%, 15%, according to Canvas defaults.

Missing more than 60% of the classes with an invalid reason implies an automatic failed module. (But in this case if you submit all of the assignments, you can pass but with a grade of at most 4.)

Feedback

Apart from grading in Canvas, I will write an email to each student about their results in detail, including corrections of their mistakes in the homework sheets.

Deadline

The 1st one (Homework 1-2) has to be submitted by the time of the 3rd session, the 2nd one (Homework 3-4) has to be submitted by the time of the 5th session, and so on, the last one (Homework 7-8) has to be submitted one week after the last session of the course.

Disciplinary Guidelines

Submitting coursework late is a minor problem, - only hinders you in understanding the course material. It only matters at the end of the course because there is a grading deadline, a few days after the end of the term.

In case of bullying fellow students and plagiarising, copying a fellow student's work, Milestone disciplinary actions apply. Plagiarism is a major problem - copying some or all exercises from someone else: usually it becomes obvious when I find the same mistake that I have seen before, then reading the whole text gives more evidence. I give zero points for these exercises, and you also lose any bonus points I would give otherwise.

Additional reading

- Course notes 1 - 8;
- Grimmett - Stirzaker: Probability and Random Processes;
- Grimmett - Stirzaker: One Thousand Exercises in Probability.